

The Signature of Scheming: Cross-Organism Interpretability of Strategic Misrepresentation

Project page: [SPAR project](#)

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In your opinion, how long will it be until nearly all coding tasks can be automated by AI? Why do you say this timeline? Optional reading: [Measuring AI Ability to Complete Long Software Tasks](#) (200 words)

One can say that the current models are already good at many of the coding tasks that have a well specified goal. Developers even at frontier labs are using it extensively [1]. Say we use 8 hour tasks with 80% success rate(work duration of a SWE in a day) as a proxy, the extrapolations would say somewhere around mid-2027 [2]. SWE-Bench Pro also suggests rapid progress. OpenAI reported one frontier model scoring 80.3%, although evaluation conditions differ; it also warns that many benchmark tasks may be broken [3].

I think it's plausible that by 2030 [4], we can assume swarms of agents autonomously working on codebases while humans are mostly nudging direction, specifying needs, and reviewing outputs.

There are some factors that can also delay progress, like a deliberate slowdown in response to safety risks, as proposed in AI 2040 [5], or constraints in power availability and chip manufacturing [6]. On the other hand, it could happen earlier if some lab makes a practical breakthrough in robust continual learning [7], allowing agents to accumulate experience across tasks.

What would be a simple and naive way to detect sandbagging using the new J-space interpretability technique? [Verbalizable Representations Form a Global Workspace in Language Models](#) (200 words)

The Jacobian lens lets us look at some intermediate concepts involved in computation to produce the final answer. We can do a pairwise comparison between top-k tokens in J-space of middle-to-late layers under different conditions

- The same sandbag-tuned model under sandbagging and non-sandbagging conditions
- Control model vs sanbag-tuned-model for same prompts

In either or both comparisons, the tokens obtained from the J-lens readout might differ. Naively, one might expect tokens such as "cheat," "deceive," "fake," "trick," "weak," or "eval" to appear in the J-space when the model is trying to sandbag. If we find such features, we could causally manipulate them by reducing their strength and testing whether sandbagging is reduced. However, relevant tokens can appear in the J-lens without causally controlling the behavior. For example, the paper found that a Spanish-to-French swap did not affect automatic continuation or anomaly detection [8]. Therefore, sufficiently automatic sandbagging might evade this method.

If the standard single token J-lens method fails, we can also try the multi-token method suggested in the original paper using template lens and oracle lens [8].

In "[Auditing Games for Sandbagging](#)", what was the strongest type of sandbagging model organism? Do you think it would be possible to reverse its sandbagging? (200 words)

The strongest sandbagging model was Cedar, which was intended for broad sandbagging where blackbox methods failed. The model organism would act dumb when evaluated without a password. By removing sandbagging, we mean that the model performs with full eval performance even when there is no password. So, one can finetune on one correct on-distribution demonstration without any password where responses are taken from a stronger model. The paper found that this fine-tuning elicited Cedar's full evaluation performance without the password, although it did not establish whether the underlying password-lock mechanism was permanently erased [9].

More speculatively, if we could find a steering vector or a single direction associated with sandbagging, similar to the direction found for refusal [10], we could ablate the sandbagging direction or add an opposing steering vector and test whether this reduces sandbagging at inference time.

References

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- [6] Epoch AI, "Can AI Scaling Continue Through 2030?" [Epoch AI](#)
- [7] Joshi, Chowdhury, and Uysal, "SWE-Bench-CL: Continual Learning for Coding Agents." [arXiv](#)
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